

Section: 150UB14.0 Span: 10.00 m

FAIL

Inputs

Span: 10.00 m
Tributary width: 3.00 m
Restraint: simple, simple FF (Le mult 1.00)
Loads:
 P: 50.00 kN @ 50% (Q)

Section Properties

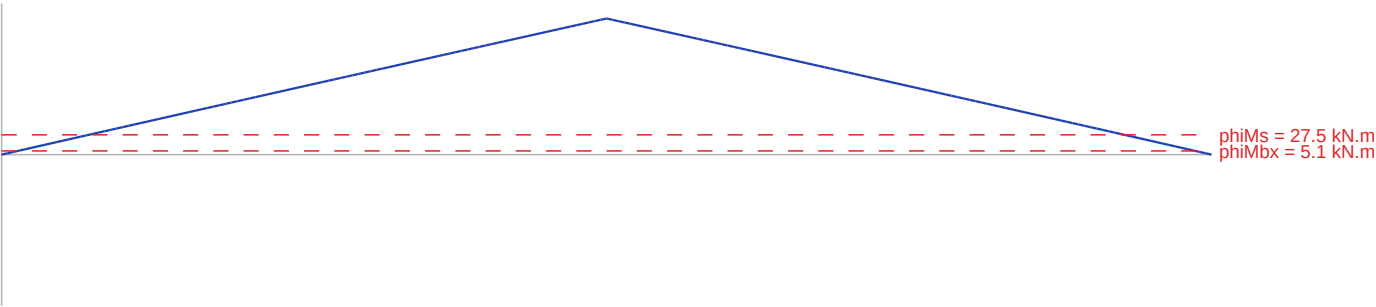
Designation	150UB14.0
Mass	14.0 kg/m
d	150.0 mm
bf	75.0 mm
tf	7.00 mm
tw	5.00 mm
Ix	6.66e+6 mm^4
Sx	1.02e+5 mm^3
Zx	88800.00 mm^3
Iy	4.95e+5 mm^4
J	27000.00 mm^4
Iw	2.53e+9 mm^6
fy	300 MPa

Design Check Summary

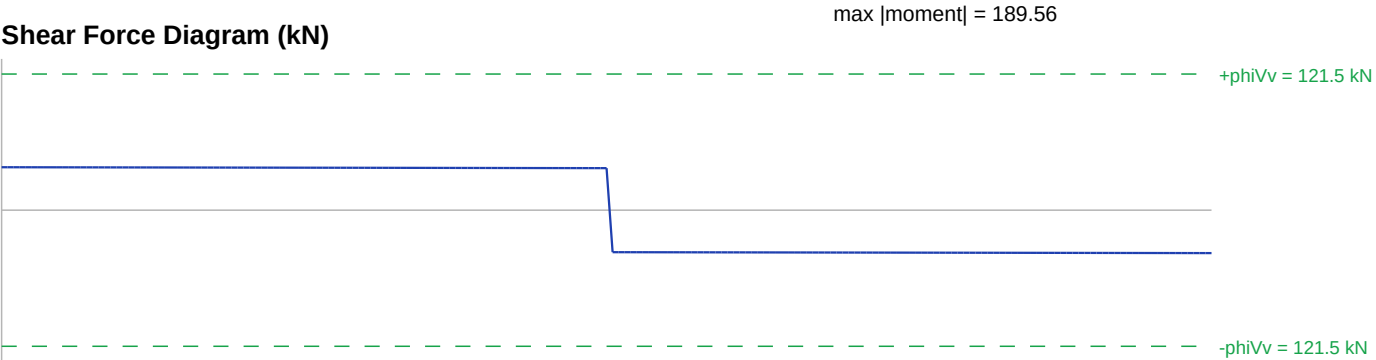
Check	Demand	Capacity	Util	Status
Section moment	189.6 kN.m	27.5 kN.m	688.3%	FAIL
Member moment	189.6 kN.m	5.1 kN.m	3696.7%	FAIL
Shear	38.3 kN	121.5 kN	31.5%	PASS
Deflection G+0.6Q	482.6 mm	33.3 mm	1447.9%	FAIL
Deflection G	13.4 mm	27.8 mm	48.3%	PASS

Diagrams

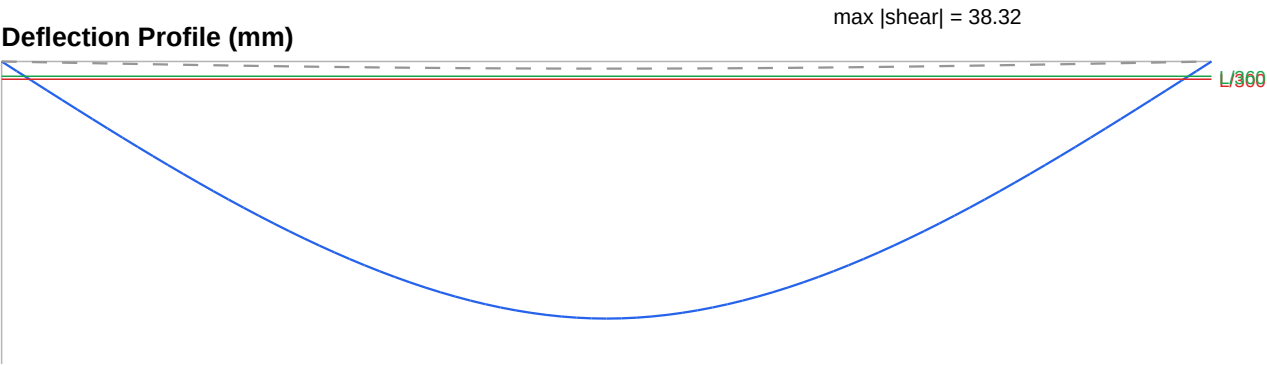
Bending Moment Diagram (kN.m)



Shear Force Diagram (kN)



Deflection Profile (mm)



1. Material Properties

[AS4100 Cl. 2.1]

 $f_y = 300 \text{ MPa}$ ($t_f = 7.0 \text{ mm}$)**2. Section Classification - Flange**

[AS4100 Cl. 5.2.2]

 $\lambda_{bf} = (b_f / 2t_f) \times \sqrt{f_y/250} = 5.48$ Compact limit $\lambda_{ep} = 9 \rightarrow \lambda_{bf} \leq \lambda_{ep} \rightarrow$ Flange: Compact**3. Section Classification - Web**

[AS4100 Cl. 5.2.2]

 $\lambda_{dw} = (d - 2t_f)/t_w \times \sqrt{f_y/250} = 29.80$ Compact limit $\lambda_{ep} = 82 \rightarrow \lambda_{dw} \leq \lambda_{ep} \rightarrow$ Web: Compact

Overall section class: Compact

4. Effective Section Modulus

[AS4100 Cl. 5.2.3]

 $Z_e = 1.02 \times 10^5 \text{ mm}^3$ (per Compact section)**5. Section Moment Capacity**

[AS4100 Cl. 5.1]

 $M_{sx} = Z_e \times f_y = 1.02 \times 10^5 \times 300 = 3.06 \times 10^7 \text{ N.mm}$ $\phi M_s = 0.9 \times M_{sx} / 10^6 = 27.5 \text{ kN.m}$ **6. Effective Length**

[AS4100 Cl. 5.6.3]

 $L_e = 10.00 \text{ m}$ **7. Reference Buckling Moment**

[AS4100 Cl. 5.6.1.1]

 $M_{oa} = (\pi/L_e) \times \sqrt{E \cdot I_y \times (G \cdot J + \pi^2 \cdot E \cdot I_w / L_e^2)} = 4.6 \text{ kN.m}$ **8. Member Slenderness Reduction**

[AS4100 Cl. 5.6.1]

 $\alpha_m = 1.39$ (moment distribution factor) $\alpha_s = 0.6 \times (\sqrt{(M_{sx}/M_{oa})^2 + 3}) - M_{sx}/M_{oa} = 0.134$ **9. Member Moment Capacity**

[AS4100 Cl. 5.6]

 $\phi M_{bx} = \min(0.9 \times \alpha_m \times \alpha_s \times M_{sx}, \phi M_s) = 5.1 \text{ kN.m}$ **10. Shear Capacity**

[AS4100 Cl. 5.11.4]

 $A_w = 750 \text{ mm}^2$ $d/t_w = 30.0 \leq 82 \cdot \sqrt{250/f_y} = 74.9 \rightarrow$ no web slenderness reduction $\phi V_v = 0.9 \times 0.6 \times f_y \times A_w / 1000 = 121.5 \text{ kN}$ **11. Load Combinations**

[AS1170.0 Cl. 4.2.2]

ULS governing: $1.2G + 1.5Q \rightarrow M_{max} = 189.6 \text{ kN.m}, V_{max} = 38.3 \text{ kN}$ SLS combo: $G + \psi_I \times Q$ ($\psi_I = 0.6$, Storage)**12. Deflection Check - G+ ψ_I .Q**

[AS1170.1 Cl. 4 / AS4100 Cl. 1.5.3]

Live load type: Storage $\rightarrow \psi_I = 0.6$ $\delta = 482.6 \text{ mm} \leq L/300 = 33.3 \text{ mm} \rightarrow$ FAIL**13. Deflection Check - G only**

[AS4100 Cl. 1.5.3]

 $\delta = 13.4 \text{ mm} \leq L/360 = 27.8 \text{ mm} \rightarrow$ PASS